# Detecting departure and destination aerodromes from ADS-B

**Version 5 — why arrivals are hard, and what recovers part of them**

EUROCONTROL Performance Review Unit

2026-08-09

## Table of contents



## Summary

Version 4 established that arrivals are harder than departures on every measure and recommended not changing the arrivals algorithm. It did not say *why*, and "we do not know why" is a poor basis for leaving a quarter of the data unlabelled. This version takes the arrival deficit apart.

**Three findings.**

**The arrival deficit is silence, not error.** Of 87,152 in-area arrivals, 3,011 are answered wrongly and **21,009 are not answered at all**. Version 4's emphasis on misnaming — built on a vivid case, Bilbao — described 12.54% of the problem.

**Most of that silence is ours, but it is not cheap to break.** 13,300 flights had a usable candidate that the abstention refused; only 7,709 had nothing at all. So the recoverable share is nearly twice the permanently lost one. But guessing on the refused flights scores 28.27% — four right for eleven wrong. The abstention is doing its job.

**One new signal beats that, and it is horizontal.** Whether the aircraft's course *points at* the candidate aerodrome recovers 957 correct arrivals for 546 new errors — 1.75:1. Vertical rate recovers nothing, and adding it to the bearing test makes things worse.

There is a single reason for that asymmetry, and it is the most useful thing in this paper:

> **!** Positions survive staleness; rates do not
>
> The OpenSky archive fills every second by carrying the last known position forward, and in this population roughly half the rows are repeats. A **bearing between two genuinely measured fixes is correct regardless of when they were measured**. A **rate divides by an elapsed time the timestamps get wrong** — the clock says ten minutes, the two real measurements may be ninety seconds apart.
> That one sentence explains why bearing alignment works, and why vertical rate, closing rate and the approach cone of version 4.5 all failed.

## 1 Where the arrival deficit sits

Table 1: Arrivals not correctly labelled, by class of the true aerodrome. 31,829 of 95,116 flights over 2025-06-05 to 07.

| Class | Flights | Coverage | Accuracy | Correct | Missing | Share of deficit |
|---|---|---|---|---|---|---|
| large airport | 80,683 | 76.11% | 96.27% | 59,120 | 21,563 | 67.75% |
| out of area | 7,964 | 5.12% | 37.99% | 155 | 7,809 | 24.53% |
| medium airport | 5,983 | 74.88% | 89.55% | 4,012 | 1,971 | 6.19% |
| small airport | 352 | 57.67% | 0.00% | 0 | 352 | 1.11% |
| heliport | 134 | 36.57% | 0.00% | 0 | 134 | 0.42% |

Two thirds of the deficit is at large airports and a quarter is out-of-area. This paper addresses the first; the second remains the largest untouched block and is left for a successor.

## 2 Silence, not error

"Not labelled" hides two situations that need opposite responses: **no candidate existed**, which is a receiver-coverage fact, and **a candidate existed and the rule refused it**, which is a choice. `nearest` never abstains, so its coverage is exactly the rate at which a candidate existed — no new computation is needed to separate them.

Table 2: The 87,152 in-area arrivals, decomposed. In-area means the true destination is a real aerodrome, so out-of-area flights are excluded.

| Outcome | Flights | Share |
|---|---|---|
| correctly labelled | 63,132 | 72.44% |

| Outcome | Flights | Share |
|---|---|---|
| answered, wrong | 3,011 | 3.45% |
| candidate existed, abstained | 13,300 | 15.26% |
| no candidate anywhere | 7,709 | 8.85% |

**8.85% of in-area arrivals have no candidate at all.** That is the hard ceiling: no endpoint-based method can exceed 91.15% coverage on this data, whatever rule it uses.

## 3 The abstained block is hard, not neglected

The 13,300 refused flights look like free coverage. They are not.

`endpoint` answers a subset of what `nearest` answers, and on shared flights both rank candidates by the same effective distance, so they return identical answers there. `nearest`'s *extra* correct answers are therefore exactly its score on the refused block: **3,760 of 13,300, or 28.27%.**

Answering them all would add 3,760 correct labels and 9,540 wrong ones — 0.39:1. Against the exchange rate version 4.5 argued for, that is not close. **"Relax the thresholds for arrivals" is now priced, and the price is bad.**

Why are they hard? Because of what an arrival endpoint is. The state-vector decimation study measured that an arrival's last fix carries a position **measured a median of 190 s earlier** — at approach speed, roughly eight nautical miles back up the approach. The endpoint is not where the flight ended. Position alone cannot identify an aerodrome from eight miles out on a busy terminal area, and the abstention correctly declines to try.

## 4 Bearing alignment

If position is exhausted, motion might not be. The measure is the angle between **the direction the aircraft actually travelled** over a window and **the direction of the candidate aerodrome**, seen from the far end of that window. Zero degrees means the track runs straight at it.

This involves no extrapolated position — it is an angle between two observed bearings — so the documented failures of trajectory extrapolation in version 2 and the 2025 PRC report do not bind it.

Table 3: Correct labels per error on the abstained arrival block, by window and angle gate. The block's own baseline is 0.37. Note that these are computed on in-area truth only, and so overstate the population-level trade – see the next section.

| Window | 1° | 2° | 3° | 5° | 10° | 20° | 30° |
|---|---|---|---|---|---|---|---|
| 2 min | 0.44 | 0.38 | 0.53 | 0.41 | 0.35 | 0.33 | 0.29 |
| 3 min | 0.60 | 0.49 | 0.58 | 0.46 | 0.40 | 0.36 | 0.31 |
| 5 min | 2.81 | 2.31 | 2.10 | 1.78 | 1.33 | 1.13 | 0.92 |
| 7 min | 3.14 | 2.66 | 2.52 | 2.38 | 1.75 | 1.21 | 0.96 |
| 10 min | 1.99 | 2.45 | 2.41 | 1.92 | 1.40 | 0.99 | 0.78 |
| 15 min | 2.09 | 1.97 | 1.79 | 1.52 | 1.07 | 0.69 | 0.55 |
| 20 min | 2.47 | 1.85 | 1.61 | 1.27 | 0.85 | 0.56 | 0.49 |

| Window | 1° | 2° | 3° | 5° | 10° | 20° | 30° |
|---|---|---|---|---|---|---|---|
| 30 min | 2.25 | 1.60 | 1.30 | 0.96 | 0.61 | 0.48 | 0.43 |

Three things are visible and all three make physical sense.

**Short windows carry no information.** At 2–3 minutes every cell sits near the 0.37 baseline. Over two minutes the aircraft's displacement is small against its position error, so the bearing is noise.

**It switches on at 5 minutes and peaks at 7.** A metric that were an artefact would not have an interior optimum with noise on both sides.

**The gate should tighten as the window lengthens.** At loose gates short windows win, because over half an hour everything points roughly everywhere. At 1° that reverses — a long baseline gives a more precise bearing, so a tight gate becomes meaningful rather than noise.

**Accuracy plateaus at 70–76%** however tight the gate. Roughly one arrival in four that points straight at an aerodrome lands somewhere else. Bearing alone has a ceiling, and tuning it further is close to exhausted.

## 5 The comparison that matters

The sweep above scores the abstained block with out-of-area truth excluded. That flatters it: rescuing a flight whose true destination is outside the observed area produces a wrong answer the block analysis never counted. The honest test is the whole population.

Table 4: Arrivals, all 95,116 ground-truth flights. Ratio is extra correct labels per extra wrong label against the base rule.

| Model | Coverage | Accuracy | Overall | Correct | Wrong | d correct | d wrong | Ratio |
|---|---|---|---|---|---|---|---|---|
| base (30 NM / 15,000 ft) | 69.10% | 95.63% | 66.08% | 62,856 | 2,870 | — | — | — |
| +bearing 7 min / 3 deg | 70.68% | 94.92% | 67.09% | 63,813 | 3,416 | 957 | 546 | 1.75:1 |
| +bearing 30 min / 1 deg | 73.13% | 93.66% | 68.50% | 65,151 | 4,407 | 2,295 | 1,537 | 1.49:1 |
| +bearing 7 min / 3 deg +rate | 69.60% | 95.22% | 66.27% | 63,038 | 3,163 | 182 | 293 | 0.62:1 |

Table 5: Departures, the same models. Bearing is actively harmful here.

| Model | Coverage | Accuracy | Overall | Correct | Wrong | d correct | d wrong | Ratio |
|---|---|---|---|---|---|---|---|---|
| base (30 NM / 15,000 ft) | 80.96% | 98.16% | 79.47% | 75,587 | 1,416 | — | — | — |
| +bearing 7 min / 3 deg | 82.44% | 97.01% | 79.97% | 76,068 | 2,347 | 481 | 931 | 0.52:1 |

| Model | Coverage | Accuracy | Overall | Correct | Wrong | d correct | d wrong | Ratio |
|---|---|---|---|---|---|---|---|---|
| +bearing 30 min / 1 deg | 82.58% | 96.80% | 79.94% | 76,037 | 2,510 | 450 | 1,094 | 0.41:1 |
| +bearing 7 min / 3 deg +rate | 81.94% | 97.54% | 79.92% | 76,019 | 1,921 | 432 | 505 | 0.86:1 |

**Arrivals gain.** +bearing 7 min / 3 deg adds 957 correct labels for 546 errors, 1.75:1. The wider +bearing 30 min / 1 deg adds 2,295 for 1,537, 1.49:1 — more volume, worse rate.

**Departures lose.** The same test adds 481 correct departures for 931 errors, 0.52:1. Consistent with everything this study has found: departures are not short of evidence, and adding evidence to a rule that is not short of it costs accuracy. **Bearing should not be applied to departures.**

⚠ The block-level ratios were optimistic, and this is why

The abstained-block sweep reports 2.52:1 at 7 min / 3°. The same rule on the full population gives 1.75:1. The difference is entirely the out-of-area flights the block analysis filtered out and the population cannot.
The lesson generalises: a rule evaluated on the subset it was designed for will beat its own field performance, and by a wide margin. Every ratio quoted in Section 4 should be read as an upper bound.

## 6 Why the vertical channel is dead

Vertical rate is the obvious second signal, and it does not work. Two independent measures of the same quantity — the broadcast `vert_rate`, and an ordinary least squares fit of altitude on time across the window — agree in sign at close to chance.

Table 6: Sign agreement between vertical measures on the abstained arrival block. 50% is chance.

| Window | vert_rate vs OLS | 2-point vs OLS | rows with vert_rate = 0 | rows fresher than 5 s |
|---|---|---|---|---|
| 5 min | 43.04% | 47.13% | 44.85% | 10.58% |
| 10 min | 51.88% | 59.80% | 45.29% | 28.87% |
| 20 min | 57.56% | 64.28% | 46.70% | 43.92% |

Table 7: The same agreement at 20 minutes, banded by the share of rows carrying a position fresher than 5 s. Agreement tracks freshness, which identifies the cause rather than inferring it.

| Window freshness | Flights | vert_rate vs OLS | 2-point vs OLS |
|---|---|---|---|
| <30% fresh | 11,960 | 49.76% | 55.39% |
| 30-60% | 10,451 | 59.71% | 60.43% |
| 60-90% | 11,309 | 62.50% | 77.51% |
| >=90% fresh | 1,067 | 71.79% | 61.39% |

Agreement rises with freshness, and 44.85% of rows carry a vertical rate of exactly zero — the signature of a value held over rather than measured. **There is no vertical signal to extract here in either form.**

The measured consequence is in the comparison table: adding the vertical test to the bearing test cuts the arrival gain from 957 correct to 182, and the ratio from 1.75:1 to 0.62:1. It does not merely fail to help — **it discards flights the bearing test correctly rescued.**

## 7 Recommendation

**Do not adopt the bearing test yet.** At 1.75:1 for arrivals it sits below the 2:1 exchange rate version 4.5 argued for, on the reasoning that a wrong aerodrome corrupts two movement counts while a silence corrupts one and is visible. It is the closest anything has come, and the first signal to beat the abstained block at all, but it is not over the line.

**Do not apply it to departures under any circumstances.** 0.52:1 is a clear loss.

**Do not pursue vertical rate on this data.** The measurement is not marginal; the channel is noise.

**Refine the bearing estimate before re-testing.** The current course is a two-point bearing across the window. Fitting it by regression, and gating on the stability of that fit rather than on window length, should raise the ratio — the 1° gate is currently winning despite a noisy input, not because of one. If that brings arrivals above 2:1, the case changes.

**And do out-of-area first regardless.** It is 24.53% of the arrival deficit against the 11.81% this recovers, and unlike the abstained block nobody has shown it is hard.

## 8 What is now ruled out

Recorded so it is not re-attempted:

- **Relaxing the abstention thresholds** — priced at 0.39:1 (Section 3).
- **The approach cone** — version 4.5 measured it against the rectangle at matched coverage; departures lost up to 2.55 pp of accuracy, arrivals were a wash.
- **Bucket decimation** — the state-vector study measured +55 correct arrivals per day for a change that rewrites `track_id` for every published row.
- **Vertical rate, in any form** (Section 6).
- **Trajectory extrapolation** — version 2 and the 2025 PRC report, independently.

## 9 Limitations

- **One sample.** Three days, 2025-06-05 to 07. The operating points were chosen as the argmax over 120 grid cells of that same sample, so the ratios are optimistic. The 2° and 3° columns peak cleanly at 7 minutes, which is why the *region* is credible; the specific numbers are not validated.

- **Two measurement bugs were found in this work**, both by sanity checks rather than by anything failing: `angle_between` used `%`, and Spark's remainder keeps the sign of the dividend, so bearings 20° apart measured 340° whenever the pair straddled north — the tell was a median alignment of 186° on a 0–180° scale. And the altitude slope divided an altitude change by an elapsed time drawn from different rows, because `min_by` skips nulls where `min` does not. Both are fixed; both had already produced numbers that were reported before being caught.
- **The `>=90% fresh` band reverses the freshness trend** for the 2-point measure, on about a thousand flights. Either small-sample noise or something real about very short tracks. Not resolved.
- **`nearest` and `endpoint` are assumed to answer identically where both answer**, which underpins Section 3. It follows from both ranking on the same effective distance, but it is an argument, not a measurement.

## 10 Reproducing

```
# the abstained block, its motion profile, and the angle x window sweep
python benchmarks/abstained_vertical.py --results-dir <dir>

# which vertical measure to trust, banded by freshness
python benchmarks/vertical_measure.py --results-dir <dir>

# base against the motion variants, whole population
python benchmarks/motion_model_compare.py --results-dir <dir>
```

Rendering this page needs no credentials, no database and no cluster.